

Level 1

- Let $\star$ be a binary operation on $\mathbb{Z}$ defined by $a \star b = 3a - 2b$. Compute $(2 \star (-1)) \star 4$.
- Consider the set $S = \{0, 1, 2, 3\}$ equipped with a binary operation $\odot$ given by the Cayley table below:

$\odot$	0	1	2	3
0	2	3	0	1
1	3	0	1	2
2	0	1	2	3
3	1	2	3	0

- Evaluate $1 \odot ((2 \odot 3) \odot 0)$.
 - Determine whether the operation $\odot$ is commutative by inspecting the table.
 - Is $\odot$ associative?
 - Does S have an identity element under $\odot$? If yes, determine which elements are invertible and find their respective inverse.
- In the set $\mathbb{Z}_8$ of integers modulo 8,
 - compute the expressions $3 \cdot (5 + 7)$. and $(3 \cdot 5) + (3 \cdot 7)$.
 - find the additive inverse of each element.
 - determine which elements have multiplicative inverse.

Level 2

- Define a binary operation $\diamond$ on $\mathbb{Q}$ (the set of rational numbers) by:

$$a \diamond b = a + b + ab, \quad \text{for all } a, b \in \mathbb{Q}.$$

- Determine whether $\diamond$ is commutative.
 - Determine whether $\diamond$ is associative.
 - Find the identity element of $\mathbb{Q}$ under $\diamond$.
 - Under what condition is an element $a \in \mathbb{Q}$ invertible with respect to $\diamond$?
- Let $n > 1$ be a fixed positive integer. Prove that if $[a] = [a']$ and $[b] = [b']$ in $\mathbb{Z}_n$, then

$$[a \cdot b] = [a' \cdot b'].$$

Level 3

- Let $*$ be an associative binary operation on a set M , with identity element e .
 - For $x \in M$, define the function $\lambda_x : M \rightarrow M$ as $\lambda_x(m) = x * m$, for all $m \in M$. Prove that the element $x \in M$ has a right-inverse if and only if $\lambda_x : M \rightarrow M$ is surjective.
 - For $x \in M$, define the function $\rho_x : M \rightarrow M$ as $\rho_x(m) = m * x$, for all $m \in M$. Prove that the element $x \in M$ has a left-inverse if and only if $\rho_x : M \rightarrow M$ is surjective.
 - Prove that an element $x \in M$ has a left-inverse if and only if the map $\lambda_x : M \rightarrow M$ defined in 1(a) is injective and $e \in \text{Im}(\lambda_x)$.
 - Prove that an element $x \in M$ has a right-inverse if and only if the map $\rho_x : M \rightarrow M$ defined in 1(b) is injective and $e \in \text{Im}(\rho_x)$.
- Let $n \in \mathbb{Z}$ be given. Prove that $n^3 \equiv n \pmod{6}$.